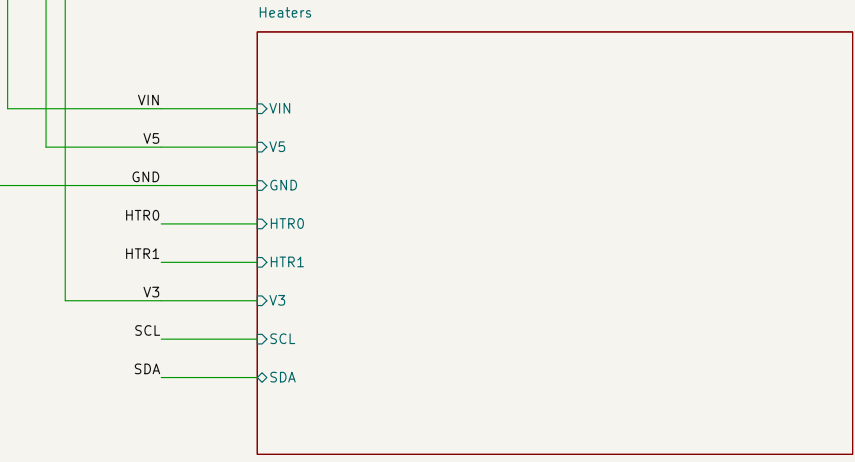
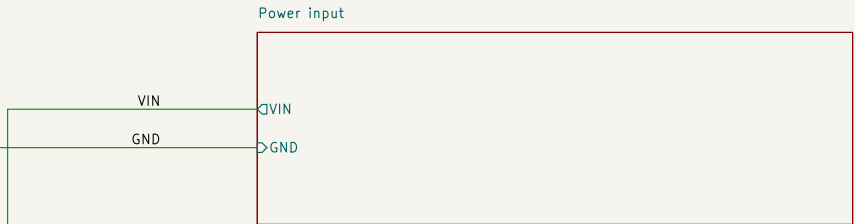


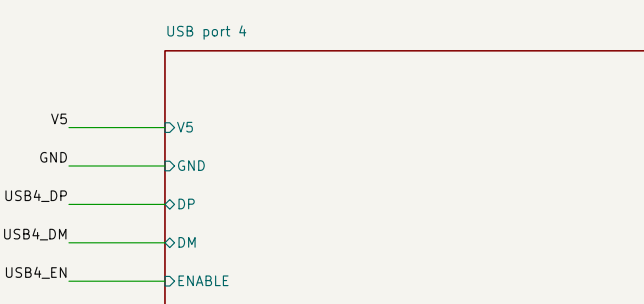
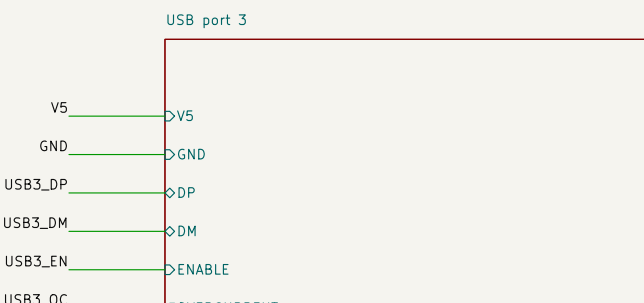
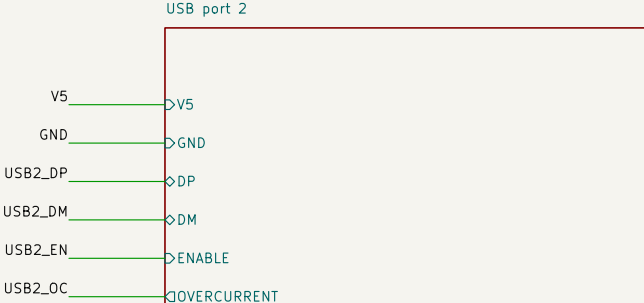
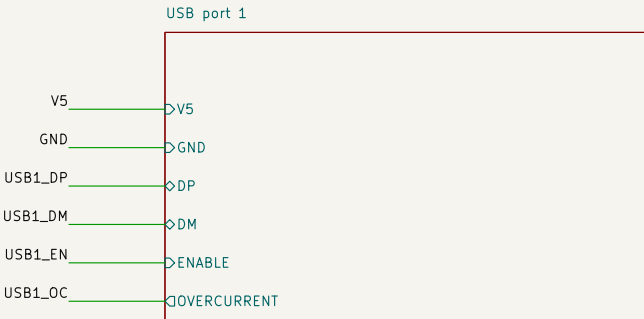
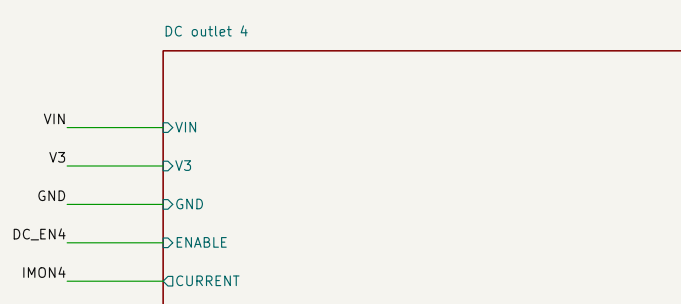
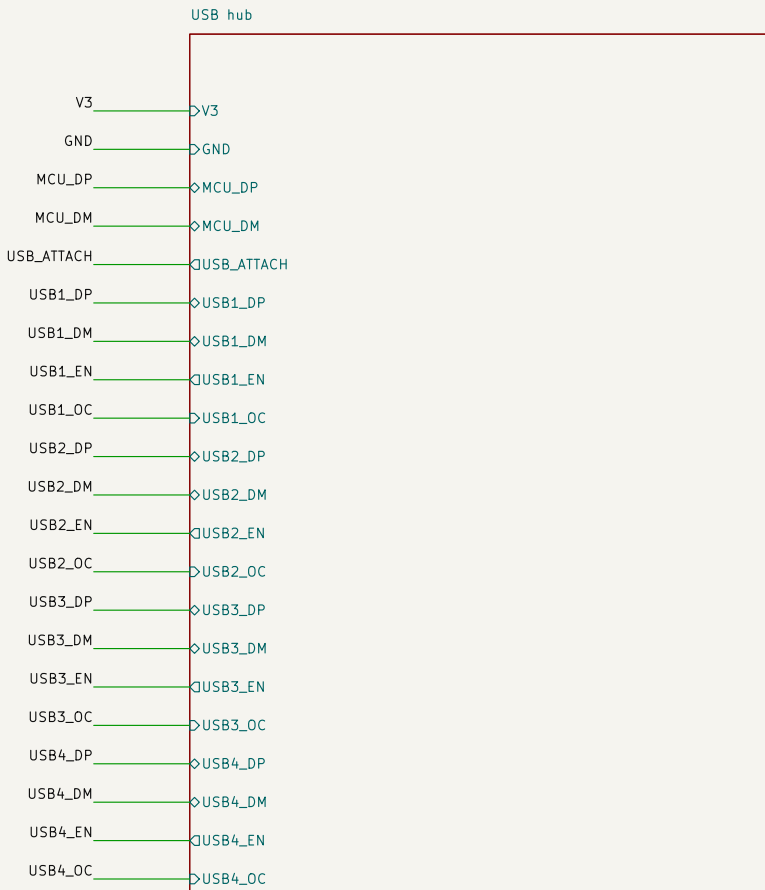
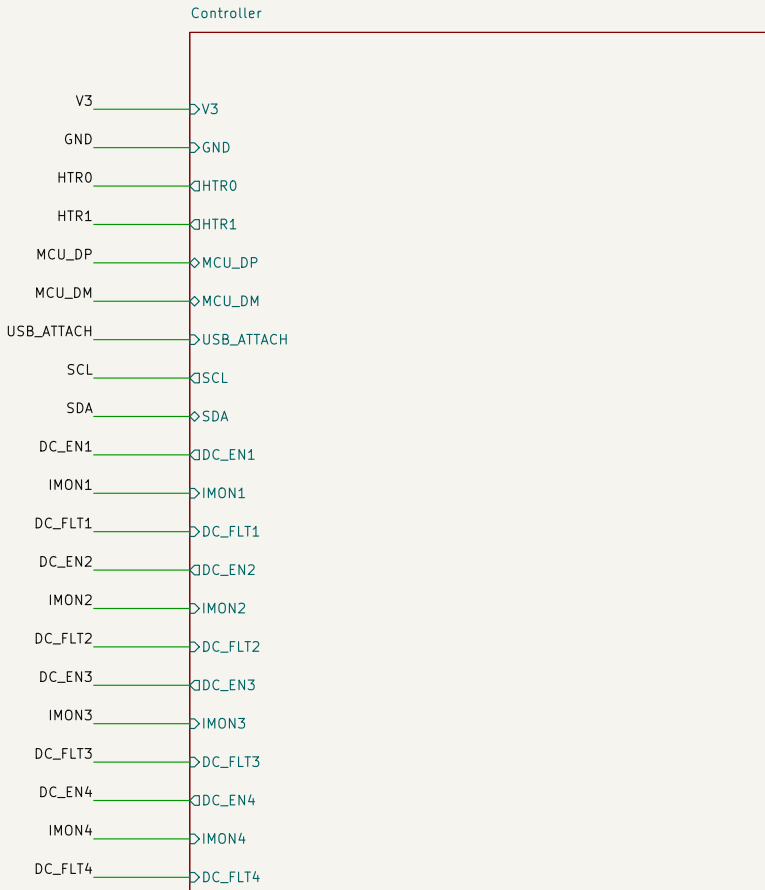
PULSARDEW HUB | 2 heaters + 4 monitored DC outlets + 4 USB 2.0 ports

PROTOTYPE FLOORPLAN – routing, full-load thermal validation and new firmware are still required.

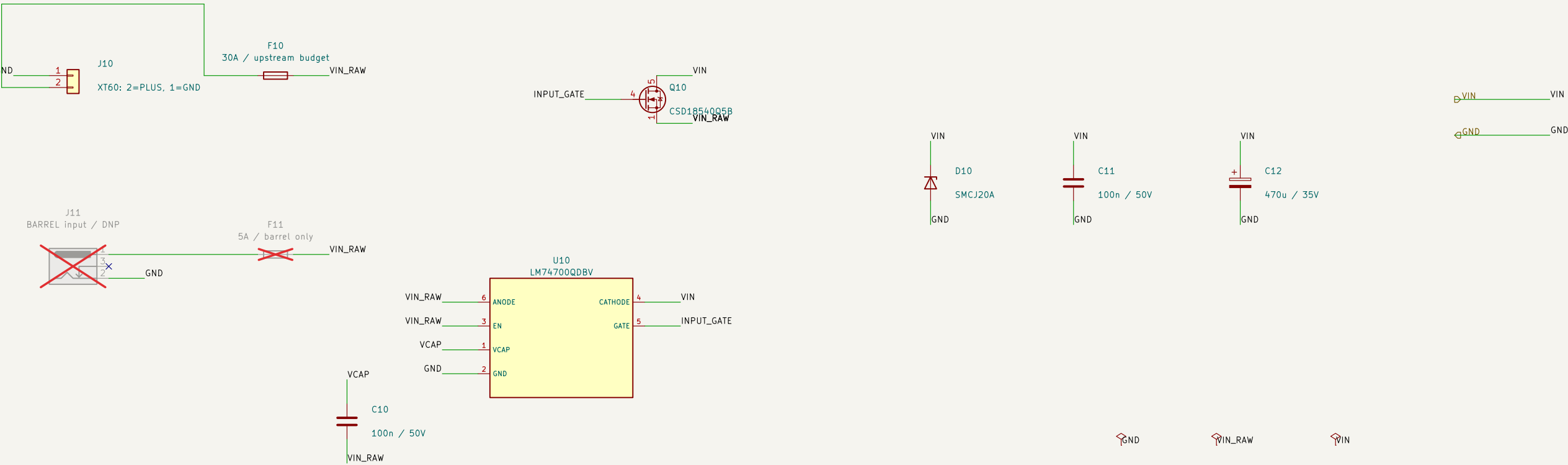


RATINGS TO VALIDATE
12–20 V nominal input
4 x 5 A DC outlet targets
2 x 5 A heater targets
25 A shared XT60 budget
3.5 A continuous barrel budget
USB: 500 mA per port
No PWM on DC outlets

Same ground throughout.
High-current and USB return paths
are separated by placement.
Do not backfeed DC outputs.

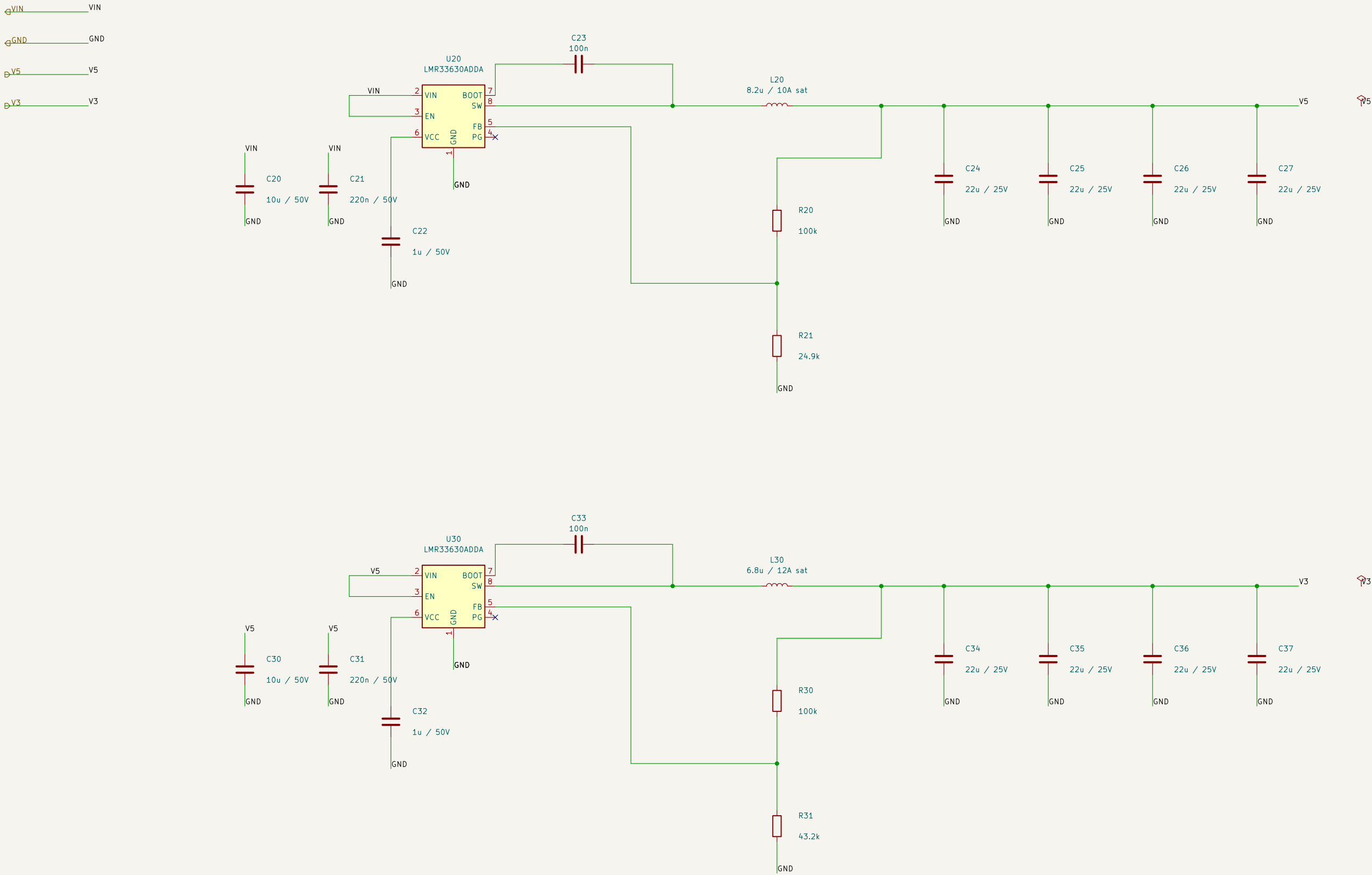


POWER INPUT | 12–20 V nominal | XT60 or barrel assembly option



FIT J10 + F10 OR J11 + F11. Do not populate both input connector paths.
XT60: provisional 25 A shared continuous budget; fuse, copper and temperature still require validation.
Barrel: 5 A absolute TOTAL ceiling; 3.5 A provisional continuous budget after fuse derating. Use a current-limited supply.
LM74700 + Q10 provides reverse-polarity / reverse-current protection. It is not an overvoltage disconnect.
Never apply more than 20 V nominal. TVS pulse capability is not a sustained overvoltage rating.

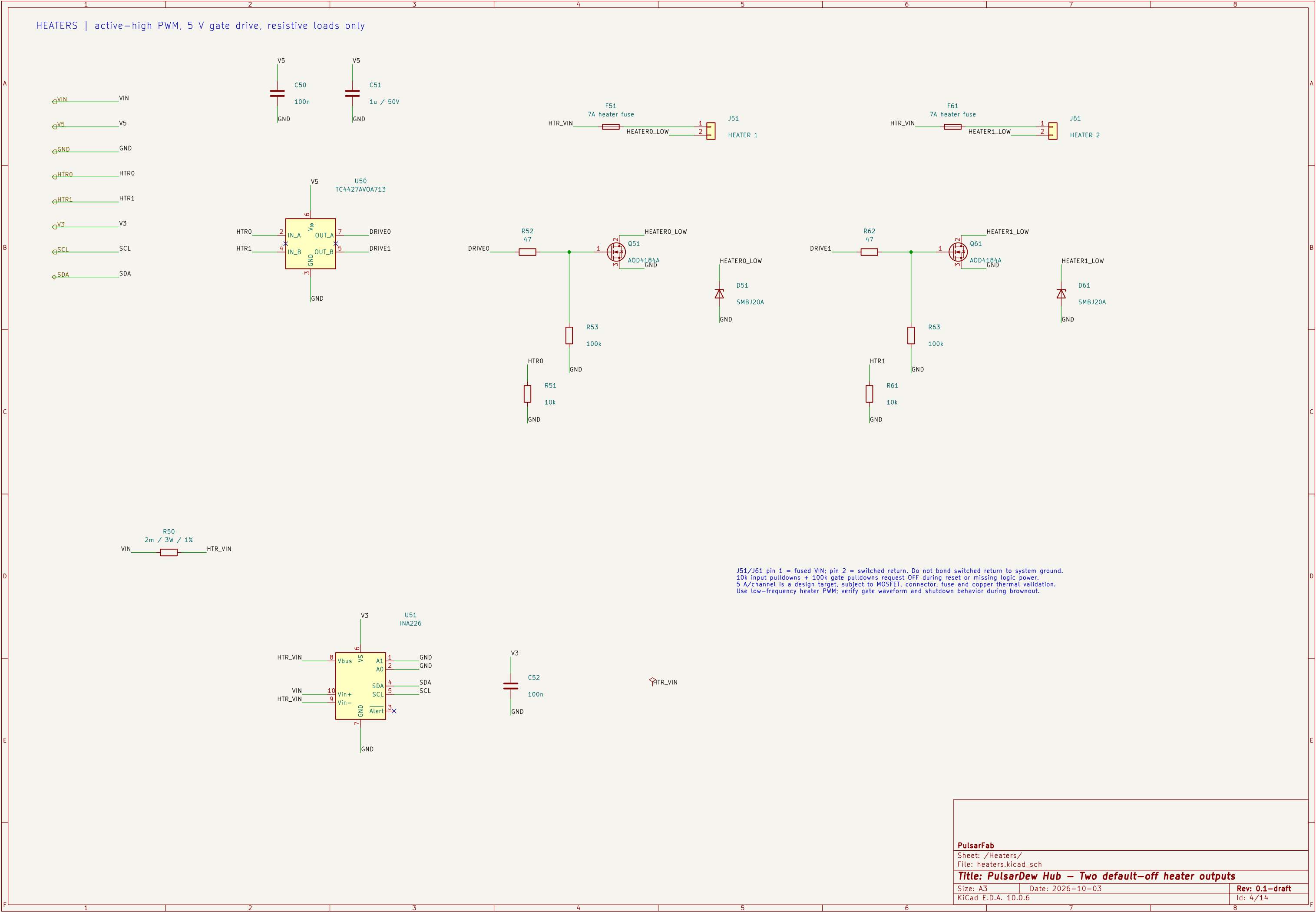
DC SUPPLIES | LMR33630ADDA, 400 kHz | 5 V / 3 A + 3.3 V logic

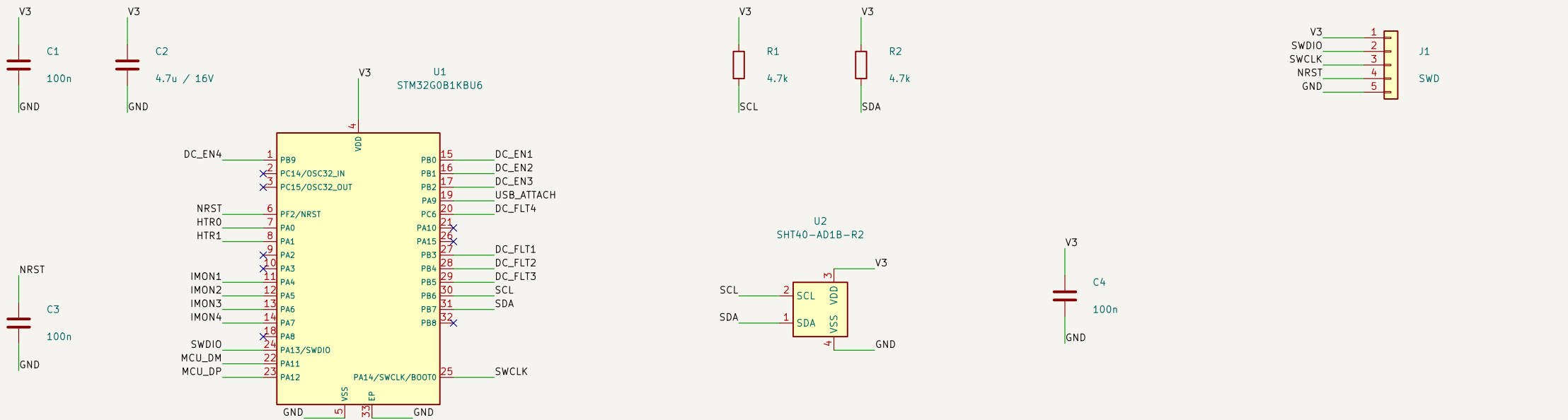


Follow TI SNVSN3F Table 9-2. 5 V uses nearest preferred 8.2 uH to the 8 uH example.
Place ceramic input/boot/VCC capacitors at the IC; keep SW copper compact and FB away from SW.
USB budget: 4 x 500 mA plus about 0.4 A for logic at full activity. Verify inductor saturation, MLCC bias, transient response and regulator temperature before routing release.

PulsarFab		
Sheet: /DC supplies/ File: supplies.kicad_sch		
Title: PulsarDew Hub – 5 V USB and 3.3 V logic supplies		
Size: A3	Date: 2026-10-03	Rev: 0.1-draft
KiCad E.D.A. 10.0.6		Id: 3/14

HEATERS | active-high PWM, 5 V gate drive, resistive loads only





V3

GND

HTR0

HTR1

MCU_DP

MCU_DM

USB_ATTACH

SCL

SDA

DC_EN1

IMON1

DC_FLT1

DC_EN2

IMON2

DC_FLT2

DC_EN3

IMON3

DC_FLT3

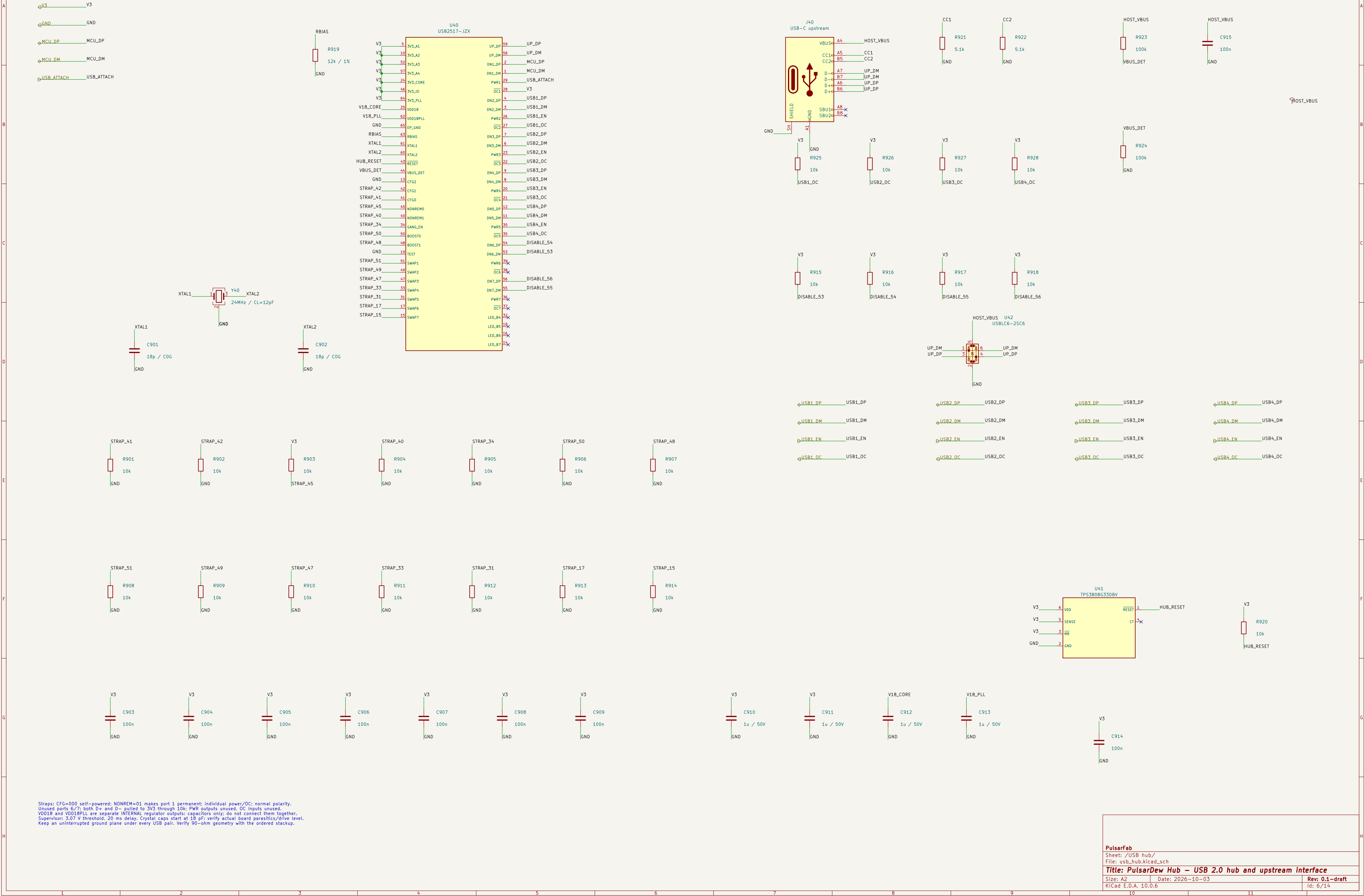
DC_EN4

IMON4

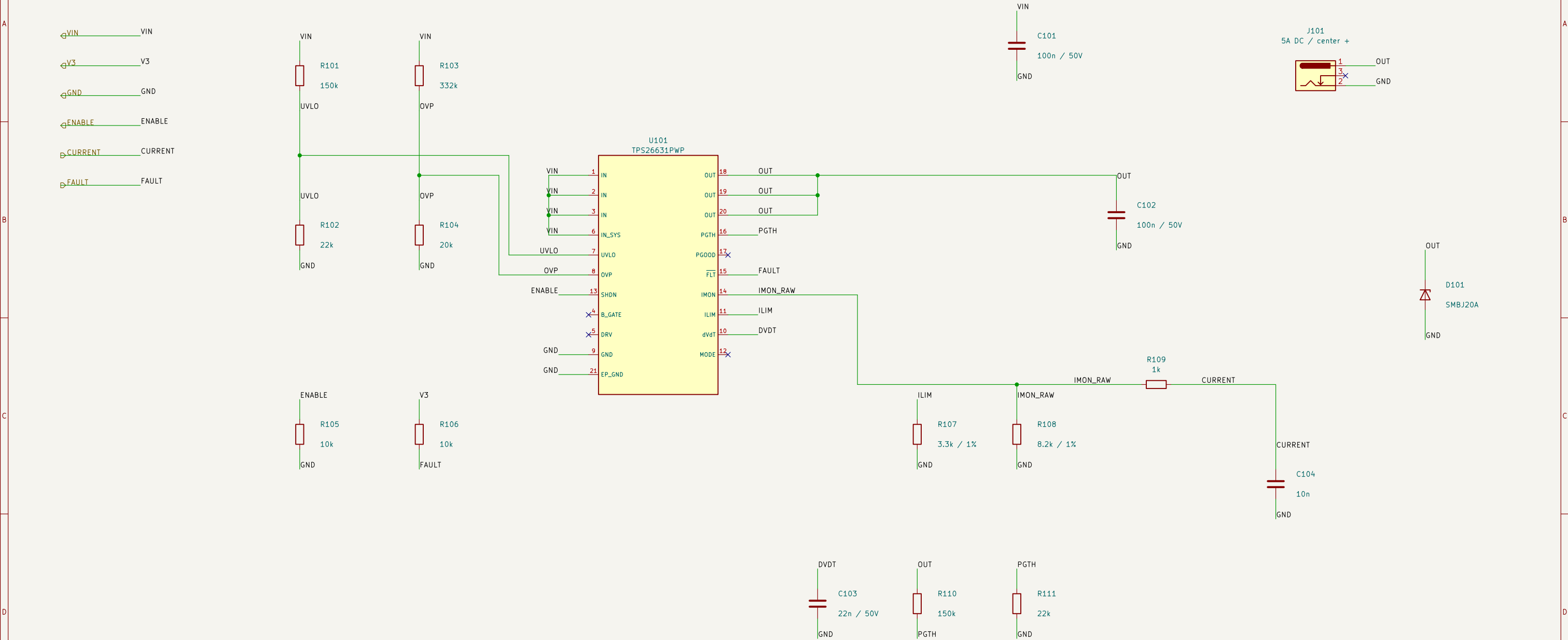
DC_FLT4

PA0/PA1: active-high TIM2 heater drive. PB0/PB1/PB2/PB9: static DC GPIO only (never PWM).
PA4..PA7: ADC IMON1..4. PB3/PB4/PB5/PC6: active-low DC faults.
PA9: internal hub port 1 enable (3.3 V). Attach USB only while high; disable UCPD dead-battery pull-down.
SHT40 at 0x44; keep physically away from power components. New firmware pin map is required.

USB HUB | USB2517 | internal port 1 + external ports 2–5; ports 6/7 disabled

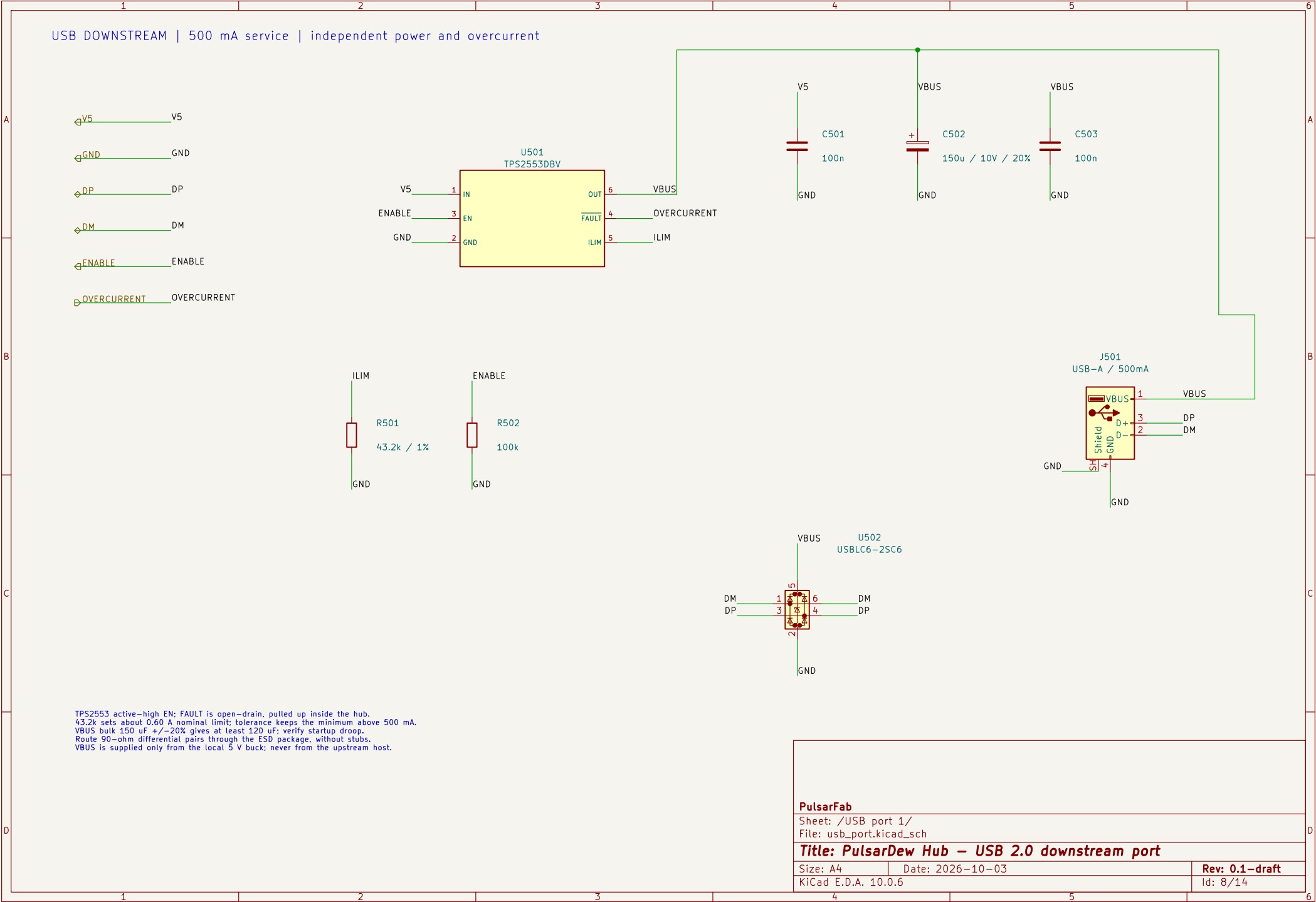


SWITCHED DC OUTLET | 5 A target | enable / current / latched fault

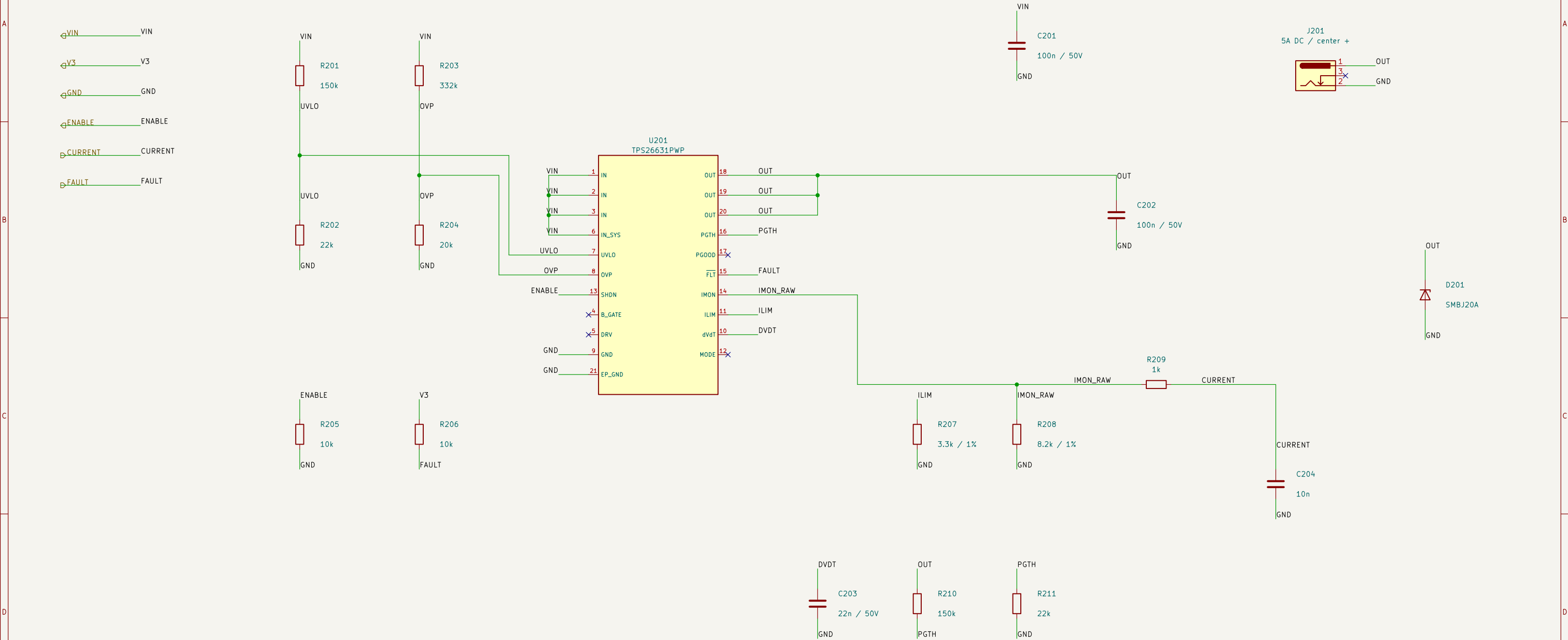


RILIM 3.3k: nominal 5.45 A (5.02..5.90 A including +/-7% IC and 1% resistor).
TPS26631 permits a 2x overload pulse before limiting/latching; this is not an instantaneous 5 A clamp.
IMON: 27.9 uA/A x 8.2k = 0.22878 V/A; +/-6% above 2 A, +/-8% at 0.6..2 A; calibrate each outlet.
UVLO -9.38 V rising / OVP -21.1 V. MODE open: latch off, reset with ENABLE low then high.
No external blocking FET; do not backfeed power into OUT. Allocate heat-spreading copper and thermal vias.
ENABLE pulldown is 10k: the SHDN internal source can be 10 uA. Plain GPIO switching only; never PWM.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

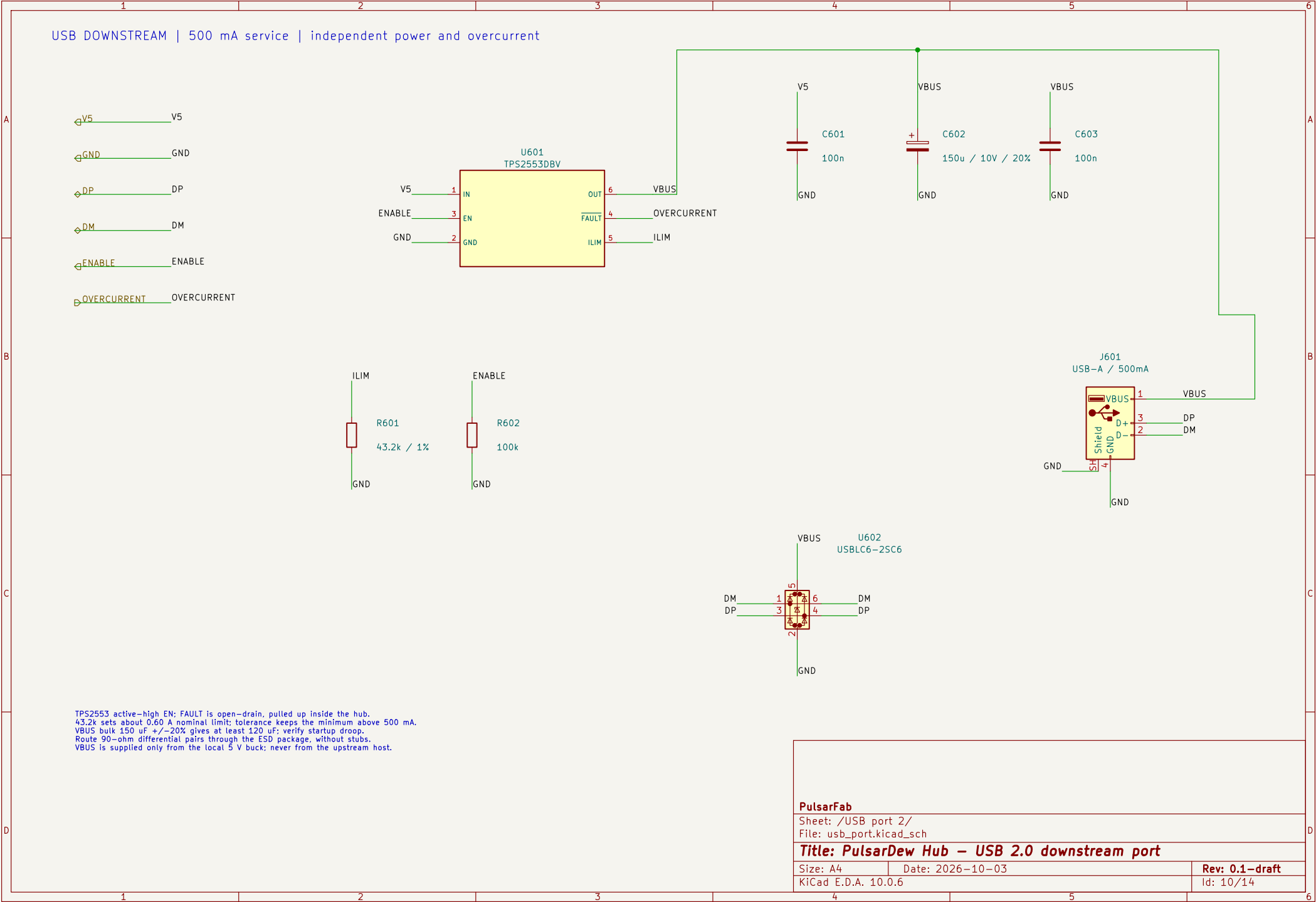


SWITCHED DC OUTLET | 5 A target | enable / current / latched fault

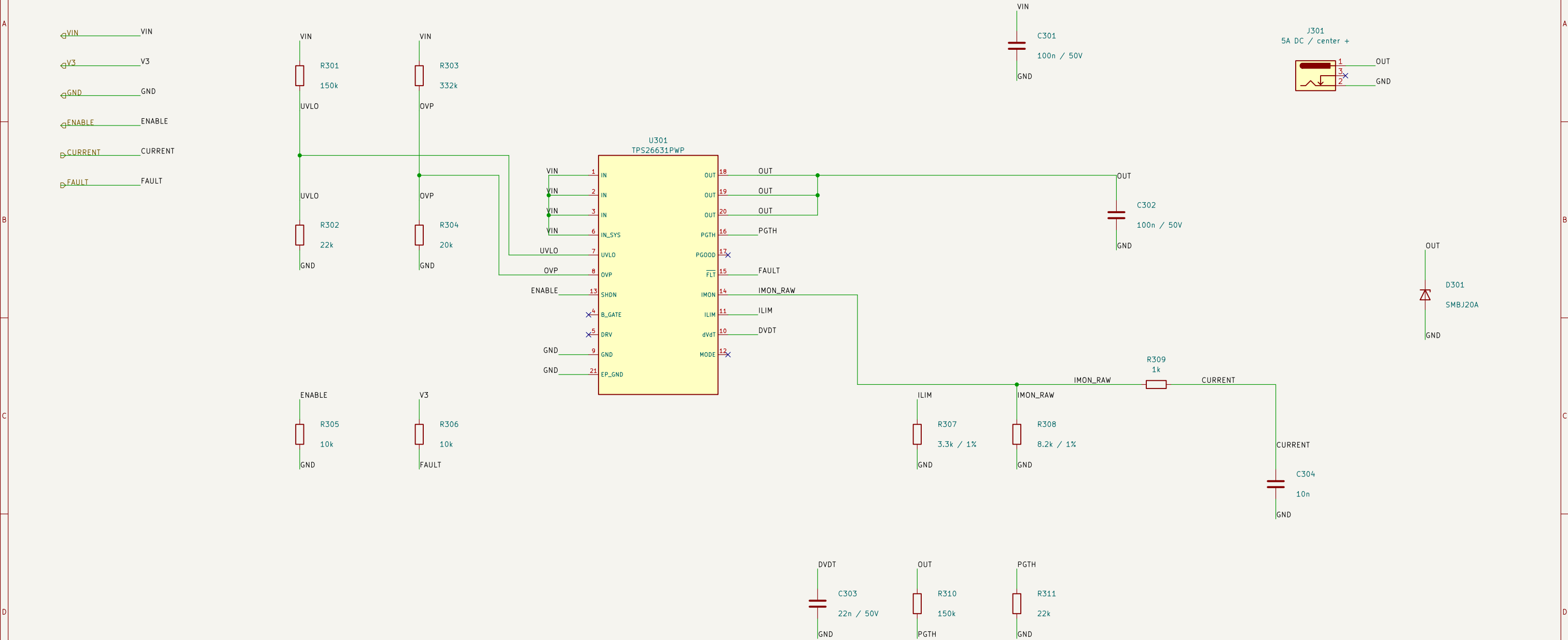


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USB DOWNSTREAM | 500 mA service | independent power and overcurrent

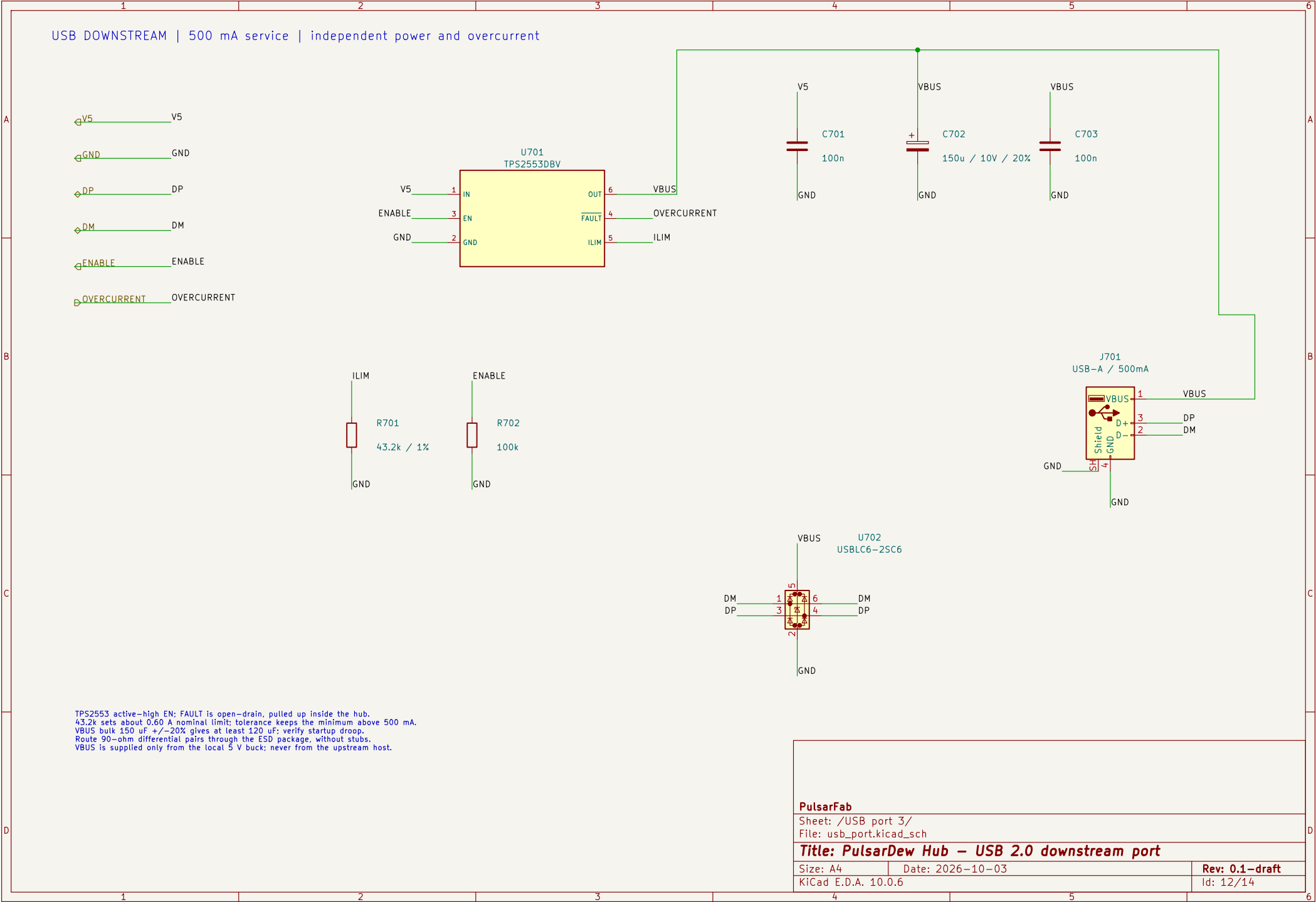


SWITCHED DC OUTLET | 5 A target | enable / current / latched fault

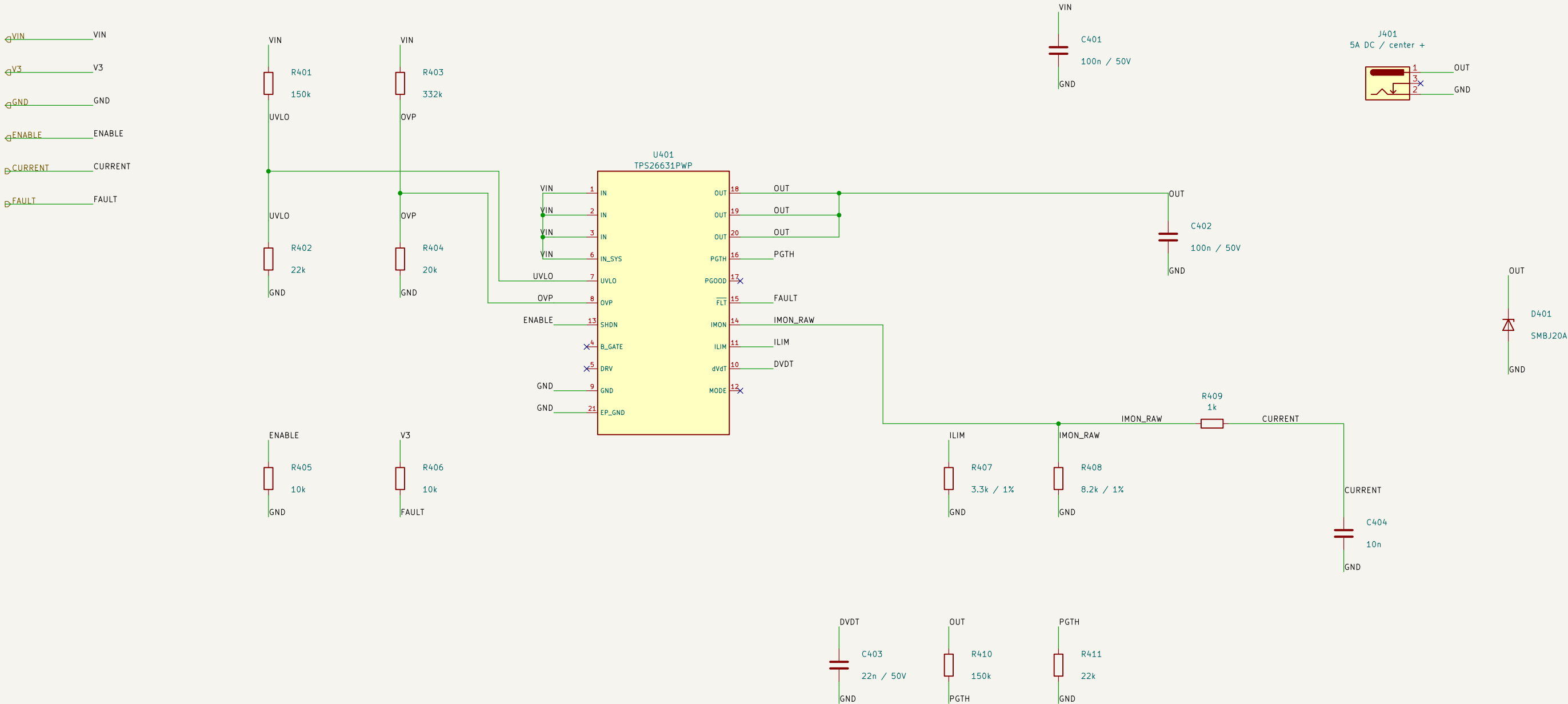


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USB DOWNSTREAM | 500 mA service | independent power and overcurrent

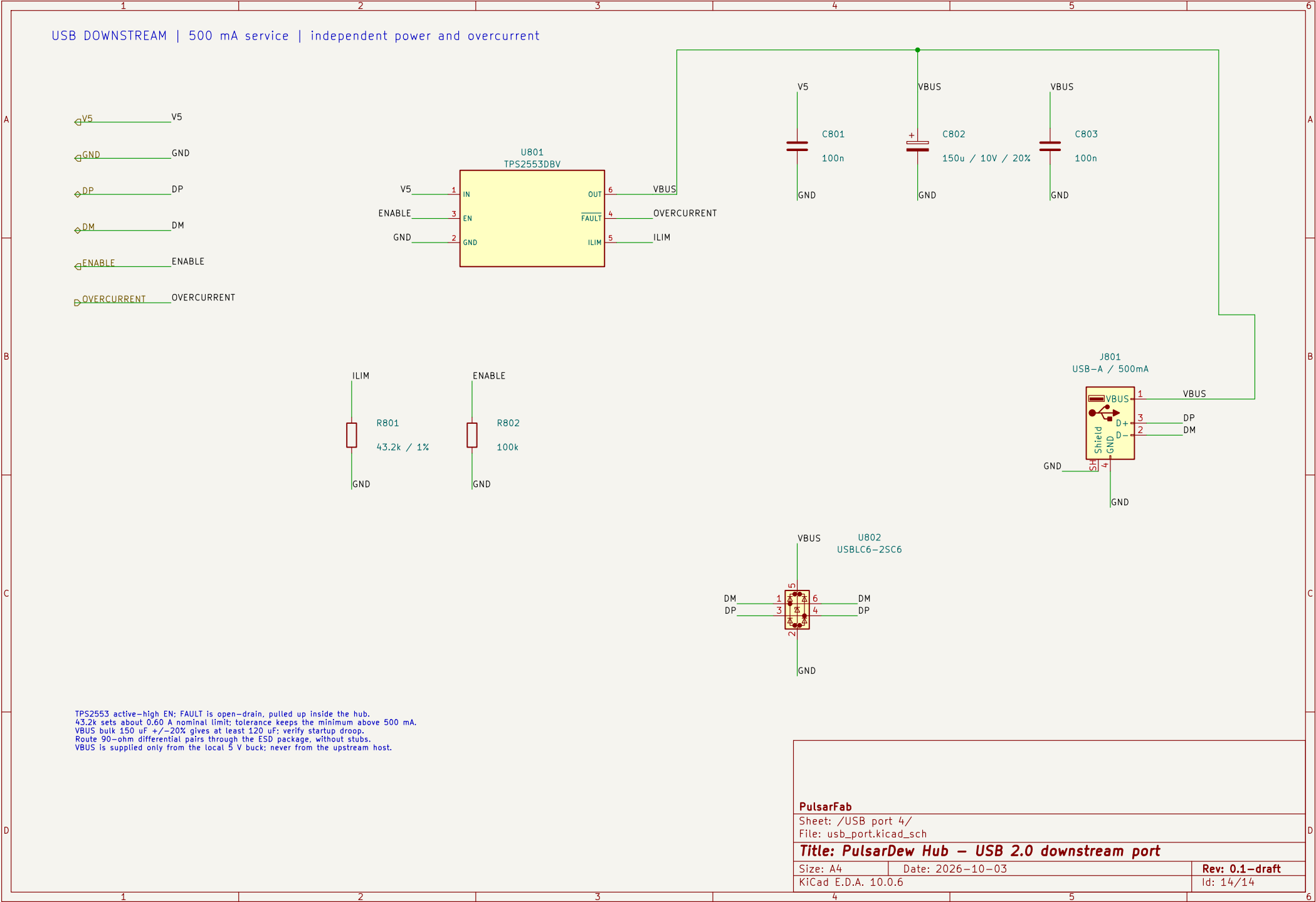


SWITCHED DC OUTLET | 5 A target | enable / current / latched fault



RILIM 3.3k: nominal 5.45 A (5.02..5.90 A including +/-7% IC and 1% resistor).
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USB DOWNSTREAM | 500 mA service | independent power and overcurrent



PulsarFab

Sheet: /USB port 4/
File: usb_port.kicad_sch

Title: PulsarDew Hub – USB 2.0 downstream port

Size: A4

Date: 2026-10-03

Rev: 0.1-draft

KiCad E.D.A. 10.0.6

Id: 14/14